

$$P(E^c) = 1 - P(E)$$

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

$$\text{De Morgan's: } \overline{(E \cup F)} = \bar{E} \cap \bar{F} \quad \binom{n}{k} = \frac{n!}{k! (n-k)!}$$

$$\overline{(E \cap F)} = \bar{E} \cup \bar{F}$$

$$\text{Classical: } \frac{\text{desired}}{\text{total}}$$

$$0 \leq P(E) \leq 1, P(\emptyset) = 0$$

$$\text{disjoint: } P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Conditional: new info shrinks sample space

$P(E|F)$: probability of E given F

$$0 \leq P(E|F) \leq 1, P(\emptyset|F) = 0$$

$$\text{disjoint: } P(E_1 \cup E_2 | F) = \sum_n P(E_n | F)$$

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

$$P(E \cap F) = P(E|F)P(F) = P(F|E)P(E)$$

Partition Rule

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

$$P(B) = \sum P(B|A_n)P(A_n)$$

→ conditional

$$P(B|E) = P(B|A_1 E)P(A_1|E) + \dots + P(B|A_n E)P(A_n|E)$$

Chain Rule: things happening in a row

$$P(A_1, A_2, A_3) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1, A_2)$$

$$\rightarrow P(A_1, A_2, \dots, A_n) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1, A_2) \cdot \dots \cdot P(A_n|A_1, A_2, \dots, A_{n-1})$$

Bayes Formula:

$$P(A_i|B) = \frac{P(B|A_i) P(A_i)}{\sum_{j=1}^n P(B|A_j) P(A_j)}$$

slipped: $P(B|A_i) \leftrightarrow P(A_i|B)$

$$(\text{prob of cause } A) = \frac{\text{prob of result given cause } A}{\text{total prob of result from all causes}}$$

Events E, F are independent if $P(EF) = P(E)P(F)$

$$P(F) \neq 0 \rightarrow P(E|F) = P(E)$$

$\rightarrow$ statistical independence: information, not physical interaction

E is ind. of $F \rightarrow$ learning that F has occurred gives no new info to help predict if E will occur

if E, F are ind., then 1) E, F' are ind.

$$\leftrightarrow E', F \text{ are ind.}$$

$$E', F' \text{ are ind.}$$

(if knowing F happened doesn't help predict E , then knowing F didn't happen doesn't help predict E either)

$$P(EF^c) = P(E)P(F^c|E)$$

disjoint events are not independent

independence of variables implies independence of events

Events A, B, C are independent if

1) $A \perp B$

2) $B \perp C$

3) $A \perp C$

4) $P(ABC) = P(A)P(B)P(C)$

$P(ABC) = P(A)P(B)P(C)$ does not imply pairwise independence

Conditional Independence

$A \perp B | C$ (conditionally independent given C) if

$$P(A|BC) = P(A|C)P(B|C) \text{ assuming } P(C) \neq 0$$

— if we know C has happened, learning B gives no info on A

$$A \perp B | C \iff P(A|BC) = P(A|C)$$

$$\hookrightarrow P(A-B) = P(A)(1-P(B))$$

$P(AB) < P(A)P(B) \rightarrow$ repulsion ($P(AB)=0 \rightarrow$ disjoint)

$P(AB) = P(A)P(B) \rightarrow$ independence

$P(AB) > P(A)P(B) \rightarrow$ attraction

geometric series: $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, |r| < 1$ Sum = $\frac{\text{1st term}}{1 - \text{ratio}}$

$$\sum_{k=0}^{n-1} ar^k = a \frac{1-r^n}{1-r}, r \neq 1$$

$$\sum_{k=1}^n ar^{k-1} = a \frac{1-r^n}{1-r}$$

Independent trials

P of getting K occurrences of E in n trials:

$$P(\text{exactly } k \text{ successes}) = \binom{n}{k} P(E)^k (1 - P(E))^{n-k}$$

$\binom{n}{k} = \binom{n}{n-k}$ if rolling 5 dice,
choosing 2 as success is the
same as choosing 3 as failures

$$\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1 \quad (\text{binomial theorem})$$

- if you add up the probability of getting

0, 1, 2, ..., n successes, it must total to 1

binomial distribution function

- probability of getting k successes in n independent trials, where each trial has the same probability of success p

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Random Variables

function that maps outcomes in a sample space to a real number

$$X: S \rightarrow \mathbb{R}$$

Let $A \subseteq \mathbb{R}$ ← event

$$\{X \in A\} = \{u \in S : X(u) \in A\}$$

u : a specific outcome

$X(u)$: scalar assigned to that outcome

CDF

CDF: cumulative distribution function

$$F_X(u) = P(X \leq u) \quad \text{for } -\infty < u < \infty$$

↑ which random var.

- tells accumulated probability $P(X \leq u)$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

General : $F_X(u) = \sum_{k \leq u} p(x=k)$

Geometric Series : $F_X(u) = \sum_{k=1}^{[u]} (1-p)^{k-1} p$

$\rightarrow F_N(u) = 1 - (1-p)^{[u]}$

CDF properties

- $0 \leq F_X(u) \leq 1$

$\rightarrow$ CDF outputs P , so value $\in [0,1]$

- if $a < b$, then $F(a) \leq F(b)$

since: $P(X \leq b) = P(X \leq a) + \underbrace{P(a < X \leq b)}_{\geq 0}$

$\rightarrow$ monotone non decreasing

if $a < b$, then $F(a) \leq F(b)$

- $F(\infty) = \lim_{u \rightarrow \infty} F(u) = 1$

$\rightarrow P(X) < \infty = 1 = 100\%$

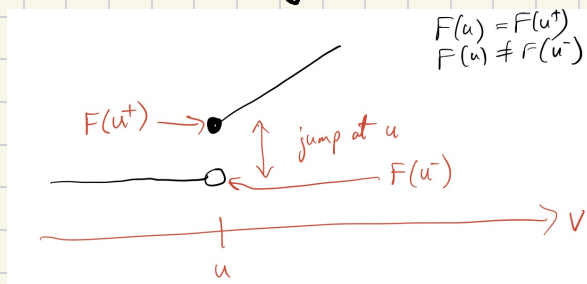
(all possible outcomes $< \infty$)

- $F(-\infty) = \lim_{u \rightarrow -\infty} F(u) = 0$

$\rightarrow P(X) < -\infty = 0 = 0\%$

(no outcomes $< -\infty$)

- $F(u)$ is right-continuous



- $P(X > u) = P(\{X \leq u\}^c) = 1 - P(X \leq u) = 1 - F(u)$
- $P(X < u) = F(u^-)$
- $P(X \geq u) = 1 - P(X < u) = 1 - F(u^-)$ ↷
- $P(X = u) = \lim_{R} F(u) - \lim_{L} F(u^-)$ $P(X > u) = 1 - P(X \leq u)$

→ $P(X = u)$ = height of jump @ u

- $P(a < X \leq b) = P(X \leq b) - P(X \leq a)$
 $= F(b) - F(a)$
- $P(a < X < b) = F(b^-) - F(a)$
- $P(a \leq X < b) = F(b^-) - F(a^-)$
- $P(a \leq X \leq b) = F(b) - F(a^-)$

If $F(u)$ cont. @ $u = a$, then no jump @ a

$$\rightarrow P(a < X \leq b) = P(a \leq X \leq b)$$

Since $F(a) = F(a^+) = F(a^-)$ (lines meet perfectly)

$$\therefore P(X = a) = 0$$

Discrete Random Variable

- can only take finite (or countably infinite) set of distinct values
- CDF graph is staircase function
- $P(X=a) > 0$

Continuous Random Variable

- can take any real number within a given range
- CDF graph is continuous (no jumps)
- $P(X=a) = 0$, only measure P in an interval

Probability Mass Function (PMF)

pmf of a discrete r.v. X :

$$P_X(u) = P(X=u)$$

p (pmf) P (probability)

- "lookup table" that maps how much P belongs to each distinct outcome
- is plugging a number u into the PMF, it tells us the P that $X=u$

- $P_X(u) \geq 0 \quad \forall u$

→ P of any specific outcome ≥ 0

- $\sum_u P_X(u) = 1$

→ sum of P of all outcomes = 1

- Uniform Discrete r.v.

→ p.m.f. has finite # of equally spaced and equal height values

→ e.g. dice roll

- Binomial r.v.

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

for $k=0, 1, 2, \dots, n$

→ counts # of successes in fixed # of trials

$$\rightarrow 1 = \sum_{k=0}^n P_X(k) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k}$$

- Poisson r.v. (with parameter λ)

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{for } k=0, 1, 2, \dots, n$$

→ counts how many times an event happens over a fixed amount of time or space

→ e.g. how many texts will I get in the next hour?

→ λ represents average rate at which things happen (4 texts per hour $\rightarrow \lambda = 4$)

$$\rightarrow \sum_{k=0}^{\infty} p_x(k) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!}$$

• Geometric r.v. (parameter

$$p_x(k) = p(1-p)^{k-1} \text{ for } k=1, 2, 3, \dots$$

→ how many trials (x) did it take to get a success?

→ parameter p is P of success on a single try

$$P(X > k) = \sum_{j=k+1}^{\infty} p_x(j) = (1-p)^k$$

$$\text{CDF: } F_x(u) = P(X \leq u) = 1 - P(X > u) = 1 - (1-p)^u$$